



University of Science
Technology, Engineering
And Mathematics
(UNSTIM)



Higher Normal School of
Technical Education
(ENSET)

**MINISTRY OF HIGHER EDUCATION AND SCIENTIFIC RESEARCH
(MERSRS)**

RESEARCH REPORT ON A SCIENTIFIC PROJECT

Specialization: Electrical Engineering

**Non-Intrusive Load Disaggregation and Intelligent Load
Shedding: A Hybrid Statistical-Symbolic Approach for
Households in Low-Resource Settings**

Prepared by:

Charbel SOSSOU BIADJA

Student completing the second year of Electrical Engineering

Independent Research and Development Project: Design and Prototyping

June 2026

Abstract

Planned power cuts and exceeded subscribed power limits deprive West African households of several hours of electricity supply every year, with a growing economic and domestic cost. This report presents an intelligent load shedder (a central microcontroller module, with a target cost below 10,000 FCFA) that decides which loads to cut no longer through fixed priority thresholds, but by real-time minimization of a four-term optimization function: financial cost, comfort, safety, and wear. The system relies on non-intrusive load monitoring (NILM, active power only) trained on the public REFIT dataset, and on a distributed architecture of three independent microcontrollers. Proteus simulation of three house profiles (4, 10, and 7 loads) shows that the optimizer drastically reduces the cascading outages typical of classic load shedders and preserves critical loads. We discuss current limitations, validation on Beninese field data, and future work: prototype, ACS712/LM35 sensors, and a local pilot with SBEE.

Table of Contents

Abstract	2
Acronyms and Abbreviations	4
List of Figures and Tables	5
Introduction and Motivation	6
Chapter 1: State of the Art and Positioning.....	7
Chapter 2: Methodology and Approach Developed	11
2.1. Genesis and Vision of the Project	12
2.1.1. Evolution of the Technical Architecture: From Relay to Simplified Hybrid System	12
2.1.1.1. From Arduino + Relay to PLC: The Quest for Reliability and Safety	12
2.1.1.2. The Challenge of Household Wiring and the Hybrid Solution	13
2.1.1.3 Final Simplification: The "High-Power" Load Shedder for the Prototype.....	13
2.2. General System Architecture.....	13
2.2.1. Modeling of Electrical Loads (RLC)	14
2.2.2. Typical Calculations for the Modeling	14
3.1. The AI Revolution: Beyond Static Priorities (The Dynamic Formula)	14
3.2. The Simulation Architecture under Proteus	16
3.3. The Load Recognition Module (NILM)	17
3.4. Machine Learning Experiments	18
Chapter 3: Results Obtained	21
4.1. Simulation Results for the Three Houses	22
House 1: The Budget House (4 loads)	22
House 2: The Large House (10 loads)	22
House 3: The Mixed House (7 loads)	23
4.2. Training Results on the REFIT Dataset.....	24
4.3.1. Choice of Base Model: From Linear Regression to Random Forest.....	24
4.3.2. Contribution of the Symbolic Layer: The Soft-Boost	25
4.3.3. Validation on 2 Million Rows.....	27
4.3.4. Summary of Lessons Learned.....	28
Chapter 4: Analysis and Evaluation of the Approach	29
5.1. Observed Performance	30
4.2. Limitations and Constraints	30
Chapter 5: Comparison with Existing Approaches.....	31
Chapter 6: Discussion and Perspectives	32
6.1. Contributions of the Hybrid Approach.....	32
6.2. Adaptation to the Beninese Context.....	32
6.3. Perspectives and Future Work.....	32
Conclusion	34
References	35

Acronyms and Abbreviations

ML: Machine Learning

AI: Artificial Intelligence

NILM: Non-Intrusive Load Monitoring

RLC: Resistance-Inductance-Capacitance (equivalent circuit)

LCD: Liquid Crystal Display

FCFA: CFA Franc — currency of the CFA zone

SBEE: Société Béninoise d'Énergie Électrique (Benin Electricity Company)

List of Figures and Tables

Figures

Diagram 1: Proteus capture of the load modeling

Diagram 2: Simulation of a three-phase source with a voltmeter

Diagram 3: Microcontroller wiring for House 2

Diagram 4: Simulation of the S function on the water heater

Diagram 4: Shedding of the air conditioner (House 1)

Diagram 5: Shedding of the microwave (House 2)

Diagram 6: Shedding of the washing machine (House 3)

Diagram 7: Performance comparison: Random Forest and Soft-Boost

Diagram 9: Washing machine prediction on 2M rows

Tables

Table 1: Overview of the authors and the link with our project

Table 2: Theses on smart-meter-based load shedding

Table 3: Personal contribution

Table 4: Load modeling

Table 5: Examples of appliances and characteristics used

Table 6: Model used for training

Table 7: Comparative performance of linear regression and random forest

Table 8: Comparison of the different versions of the symbolic layer

Table 8: Performance of the hybrid system on 2 million rows

Table 9: Performance observed during the Proteus simulation

Table 10: Comparison of the different aspects of the project

Introduction and Motivation

In West Africa, and particularly in the Republic of Benin, the domestic electricity supply is subject to numerous disturbances, among them: planned outages imposed by the grid operator during periods of high demand, and domestic micro-outages caused by exceeding the subscribed power limit. In March 2026, SBEE and SBPE acknowledged, in a joint statement, repeated disruptions across the entire national territory linked to technical constraints on regional interconnections (Bénin Intelligent, 24 March 2026). In June 2026, SBEE again scheduled a wave of planned outages affecting eight cities simultaneously (Ouidah, Dogbo, Lokossa, Bohicon, Parakou, Kandi, Dassa, Djougou), with up to seven consecutive hours of interruption in Djougou (La Nouvelle Tribune, 30 June 2026, Benin). The classic response to this is the priority-threshold load shedder, which cuts loads "from the bottom up" without regard to the cost of lost comfort, the mechanical wear induced by repeated short cycling of household appliances, or the tariff in effect.

This project proposes an alternative: a decision-based load shedder. Rather than responding to a classic priority threshold with priority and non-priority logic, it solves a small combinatorial optimization at every instant, whose optimization function explicitly expresses the four dimensions of the problem. In doing so, it follows on from academic research on non-intrusive load monitoring (NILM, Hart 1985) and home demand-side management (Chen et al. 2019; Wu et al. 2024), adapting it to the context of Beninese grids and to an affordable cost threshold.

Chapter 1: State of the Art and Positioning

Before turning to the subject proper, it is worth reviewing recent research and positioning our contribution accordingly. This chapter is therefore devoted to that review and to our positioning. Its purpose is to situate the project relative to existing academic work and to add our contribution to it.

1.1. State of the Art

1.1.1. The Foundational Work (Hart, MIT, 1985)

The decomposition of an aggregate signal into individual appliance contributions traces back to the foundational work of George Hart at MIT (1985), who showed that a single measurement point (the meter) is enough to identify and monitor a home's appliances, by detecting active-power steps and matching them to a dictionary of signatures. This approach underlies all modern NILM systems, including ours. Kolter and Jaakkola (2011)

formalized the approach using factorial Hidden Markov Models on the REDD dataset. More recently, work on public datasets has shown that low-frequency active power alone remains a sufficient descriptor: Shahab et al. (2024) achieve 94.6% disaggregation accuracy on REFIT, the dataset of 20 British homes from Loughborough (2014-2016).

1.1.2. HEMS Systems (Shareef, Chen, Wu)

On the subject of load control, three closely related works directly frame this project: the **Chalmers thesis on forced load shedding via the smart meter relay, the AI-based edge-computing**

Arduino smart meter of Chen et al. (Sensors, 296 citations), and the low-cost OpenADR demand-response controller built on Arduino by Wu et al. (2024), along with several theses. The table below details their contributions and the link with our project.

Table 1: Overview of the Authors and the Link with Our Project

Author(s)	Year(s)	Contribution	Link with Our Project
Shareef et al.	2018	Review of HEMS (Home Energy <i>Management Systems</i>), 584 citations	It provides the general architecture: sensor - microcontroller - AI - user
Chen et al	2019	Smart Meter Arduino with Edge <i>Computing AI</i>	Validates the use of Arduino for demand and for simulation.
Wu et al.	2024	Low-Cost Demand Response Controller on Arduino, OpenADR-Compatible	Proof that an Arduino-based system can be reliable and evaluated under real-world conditions.

Table 2: Theses on Smart-Meter-Based Load Shedding

Author / Institution	Title	Contribution
Chalmers Thesis (Sweden)	On Possibilities of Using Smart Meters for Compulsory Load Shedding	Use of the smart meter's relays (Arduino components) for load shedding.
Zambia Thesis (Arduino Mega)	Peak-load shifting utilizing load factor optimization	Validates the use of microprocessors.
UQTR Thesis (Canada)	Intelligent management of electric vehicle charging (2023)	Application of AI to household load management.
ENSIAS Thesis (Morocco)	AI for energy optimization with smart meters (2024).	Morocco's energy context is close to that of our own country, Benin.

1.1.3. Positioning of Our Project

Our project follows on from this body of work while making original contributions of its own. The table below illustrates this well.

Table 3: Personal Contribution

Component	Existing Work	Contribution
Appliance Recognition	Hart (1985), Kolter (REDD)	
Architecture	Chen (2019), Wu (2024)	Single central module per house, simulation on three (3) houses, target cost below 10,000 FCFA (prospective)
Load-Shedding Algorithm	Chalmers Thesis (meter relay)	Dynamic four-term cost function instead of fixed thresholds
Context	SCIREP Central Africa (2024)	Adaptation to the Beninese context (SBEE, 230 V, 10 A)

Chapter 2: Methodology and Approach Developed

This second chapter is devoted to the methodology and the approach developed. It is split into two parts: one for the simulation side and one for the NILM side.

2.1. Genesis and Vision of the Project

The intelligent load shedder project was born from a critical observation of electrical realities in Benin, particularly in the field. The initial objective was to design a system able to cut loads one at a time (rather than all at once) to avoid overloads, reduce energy costs, and protect appliances, all without resorting to connected smart plugs or complex wireless protocols (Wi-Fi/Bluetooth). The initial constraints were clear:

- Local context: electrical installations that are often simplified (a single circuit breaker, connections via junction boxes).
- AI innovation: integration of a compressed, local narrative AI (Edge AI) for the microcontroller, able to communicate and explain why it is cutting each appliance while leaving the final choice to the user.
- Autonomy: the device had to include a battery to keep operating during power outages and to prepare for them in advance.
- Cost: a very aggressive target selling price of under 10,000 FCFA (about €15) for the Beninese market, implying frugal engineering from the design stage onward.

2.1.1. Evolution of the Technical Architecture: From Relay to Simplified Hybrid System

The research journey saw the load shedder's architecture evolve significantly to adapt to the technical and economic constraints in the field.

2.1.1.1. From Arduino + Relay to PLC: The Quest for Reliability and Safety

Initially, the idea of using individual relays driven by an Arduino was considered. However, a comparative analysis quickly revealed the limits of this approach for large-scale deployment at a controlled cost:

- Hidden cost of relays: although a single relay is cheap, the accumulation of relays, wiring, protection devices (fuses, circuit breakers), and assembly time makes this solution more expensive and more complex than using a PLC as soon as the number of loads exceeds three or four.
- Safety and reliability: industrial PLCs are designed for robustness and safety (certifications, fire-resistant materials), unlike hand-built relay assemblies, which can pose risks of overheating or electrical arcing.
- Simplified wiring: a PLC centralizes inputs/outputs, drastically reducing wiring complexity compared with a multitude of relays. This analysis led to favoring the integration of a micro-PLC (of the ESP32-PLC type, or solutions such as NORVI/CLICK)

as the execution arm, driven by the ESP32 for the AI. This solution offered the best trade-off

between industrial robustness and programming flexibility.

2.1.1.2. The Challenge of Household Wiring and the Hybrid Solution

Field observations revealed a crucial reality: household wiring in Benin is often organized by zone (a single circuit for an entire room, feeding both light and fan), and junction boxes are often saturated. This posed a major problem for selective load shedding:

- Problem: cutting a circuit (for example, the lights) at the panel switched off every load in the room, including priority lighting.
- Hybrid solution: the architecture evolved toward a "Teacher and Student" approach: a central module (Teacher) located at the electrical panel, integrating the ESP32 (embedded AI) and controlling heavy loads (air conditioner, water heater) via contactors, and Satellite modules (Student), small compact modules (micro-relays) installed locally in junction boxes or at the ceiling-fan rheostats. These modules communicate wirelessly with the Teacher to shed specific loads (e.g., a ceiling fan) without affecting the lighting.

2.1.1.3 Final Simplification: The "High-Power" Load Shedder for the Prototype

To reach the 10,000 FCFA cost target and speed up development of the first prototype, the approach was simplified: focus solely on shedding high-power appliances:

- Rationale: modern LED lamps consume very little, and the major energy savings come from managing the big consumers (fans, air conditioners, irons, etc.).
- Simplified architecture: the system mainly consists of a central module at the electrical panel that intercepts high-power load circuits via contactors. This reduces complexity and cost by eliminating the need for complex satellite modules in the first prototype.

2.2. General System Architecture

The system is designed as a central module placed at the electrical panel. It consists of: an Arduino microcontroller (simulated in Proteus), a current measurement (simulated by the FILE generator injecting the REFIT dataset), relays controlling the loads, an I2C LCD screen for real-time display, and a buzzer for alerts.

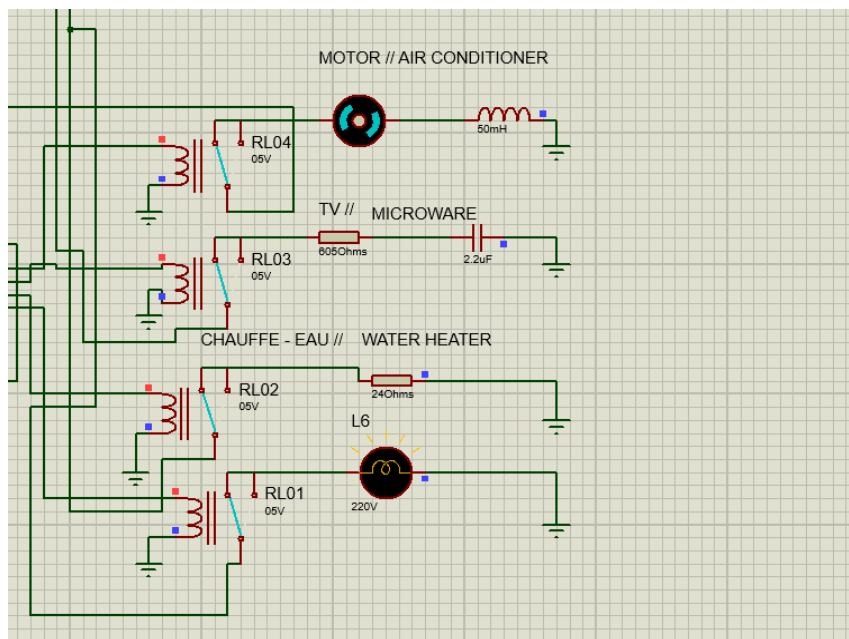
2.2.1. Modeling of Electrical Loads (RLC)

Each appliance is modeled by an equivalent electrical circuit. The R, L, C values are derived from the REFIT dataset and from the specifications of typical appliances (power P, power factor $\cos \phi$), under a voltage of 220 V and a frequency of 50 Hz. The three load types used are:

Table 4: Load Modeling

Load Type	Diagram	Example Appliance
Purely Resistive (R)	Resistance only	LED lamp, water heater, iron
Inductive (Series RL)	Resistance and inductance	Refrigerator, motor
Capacitive (Series RC)	Resistance and capacitor	Television, microwave, computer
RLC Circuit	Resistance, inductance, and capacitor	Air conditioner

Diagram 1: Proteus capture of the load modeling.



2.2.2. Typical Calculations for the Modeling

Purely resistive load (pure R):

$$R = U^2/P, \text{ which for a lamp gives } R = 220^2/10 = 4840 \, \Omega \quad \frac{U^2}{P} \quad \frac{220^2}{10}$$

Where

R: the resistance of the typical appliance

U: the voltage

P: the power of the appliance

3.1. The AI Revolution: Beyond Static Priorities (The Dynamic Formula)

One of the project's major innovations is the integration of an AI that goes beyond simple fixed-priority management. Thanks to a dynamic optimization formula, the intelligent load shedder acts as a genuine system, able to make optimal decisions in real time. The following formula, inspired by optimization principles from artificial intelligence, lets the load shedder compute a "score" for each possible on/off scenario of the appliances. The goal is to minimize this score, which represents the total cost (financial, comfort, safety, wear) over a given period t.

$$S = \int_t^{t_f} \left[\pi(t) \cdot \sum_i p_i \cdot x_{i(t)} * \sum_i \alpha_i \cdot \gamma_{i(t)} \cdot (1 - x_{i(t)}) + \gamma \cdot \max(0, \sum_i P_i \cdot x_{i(t)} - p_{max})^2 + \sum_i u_i \frac{dx^2}{d(t)^2} + \mu_i \cdot T_{i(t)} \right]$$

This formula breaks down into four main terms, each representing a key aspect of the optimization:

Term 1: The Financial Cost $(\pi(t) \cdot \sum_i p_i \cdot x_{i(t)})$

This term represents the direct cost of the electricity consumed. It takes into account:

p_i The power of appliance i,

$x_{i(t)}$ The state of appliance i at instant t (0 if on and 1 if off),

$\pi(t)$ The price of kWh at that instant. This makes it possible to factor in variable tariffs (peak/off-peak hours) or different production costs (grid power vs. generator). The AI seeks to minimize this cost by switching off appliances when the price of electricity is high.

Term 2: The Comfort / Frustration Cost $\sum_i \alpha_i \cdot \gamma_{i(t)} \cdot (1 - x_{i(t)})$

This term quantifies the discomfort felt by the user when an appliance is switched off. It introduces a notion of dynamic priority:

α_i : a static priority weight for the appliance. A lamp will have a very high weight (high frustration if cut), a ceiling fan a moderate weight.

$\gamma_{i(t)}$: an urgency factor that increases with the time the appliance remains off. The longer an appliance stays off, the more the comfort penalty grows, forcing the AI to switch it back on if possible. This term guarantees that the AI does not indefinitely cut off an appliance, even a non-priority one, and accounts for the user's need for comfort.

Term 3: The Safety / Overload Cost $\gamma \cdot \max(0, \sum_i P_i \cdot x_{i(t)} - p_{max})^2$

This term is a penalty in case the maximum allowed power (Pmax) of the installation (subscribed power or generator capacity) is exceeded: p_{max}

P_i : the total power consumed at that instant.

p_{max} : the maximum power the installation can support.

γ : a very high coefficient. The square of the overshoot makes this penalty exponential, guaranteeing that the AI will do everything it can to avoid exceeding it (and thus tripping the installation's breaker). This term ensures electrical safety and the stability of the home network.

Term 4: The Maintenance / Appliance Wear Cost $\sum i u_i \frac{dx^2}{d(t)^2} + \mu_i \cdot T i_{(t)}$

This particularly innovative term takes into account appliance lifespan and quality of service:

$\sum i u_i \frac{dx^2}{d(t)^2}$: penalizes state changes that occur too frequently (on/off cycling).

$\frac{dx^2}{d(t)^2}$ represents the speed of state change. A high weight for a fragile appliance (e.g., an air conditioner compressor) will discourage the AI from switching it off or back on too often.

$\mu_i \cdot T i_{(t)}$: penalizes operating cycles that run too long. It is also the continuous time during which the appliance has been on. A high value for an appliance sensitive to overheating (e.g., a motor) will encourage the AI to grant it rest periods. This term lets the AI manage appliance wear, optimize their lifespan, and improve quality of service by avoiding unwanted micro-outages. $T i_{(t)}$

This load-shedder formula does not merely follow fixed rules. At every instant, it evaluates the different possible combinations of switching sheddable loads on or off. For each combination, it computes the score S and chooses the configuration that minimizes this score. This dynamic process allows continuous adaptation to variations in price, consumption, comfort, and appliance state, turning the load shedder into a genuine intelligent energy optimizer.

3.2. The Simulation Architecture under Proteus

The simulation relies on a set of three houses connected to a three-phase source with one phase per house (similar to SBEE's power distribution), on three independent Arduino microcontrollers, one per house.

Each Arduino reads the input current on the analog pin (A0), computes the total power, evaluates the S function for each possible combination of states ($2^4 = 16$ combinations for house 1, $2^7 = 128$ for house 3, $2^{10} = 1024$ for house 2), selects the action that minimizes S, drives its relays on digital outputs D2 to D12, and displays the power in real time on an I2C LCD screen. A buzzer (output D7) signals critical situations.

Diagram 2: Simulation of a three-phase source with a voltmeter to measure the output voltages.

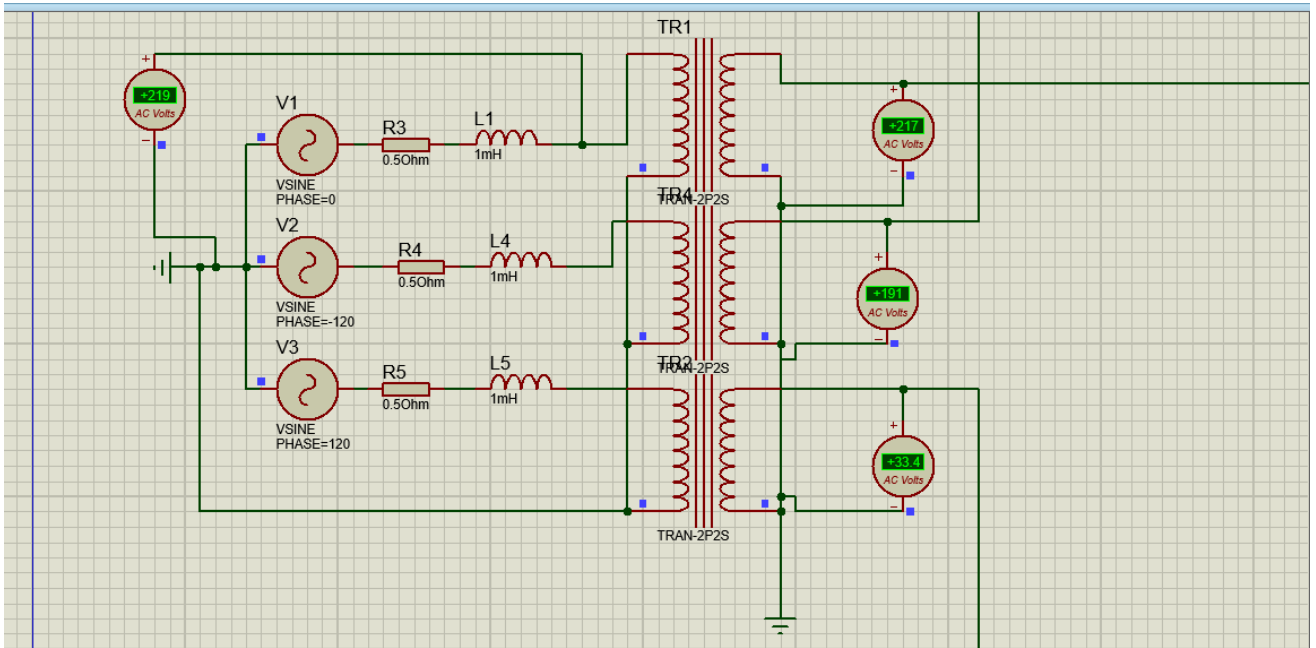
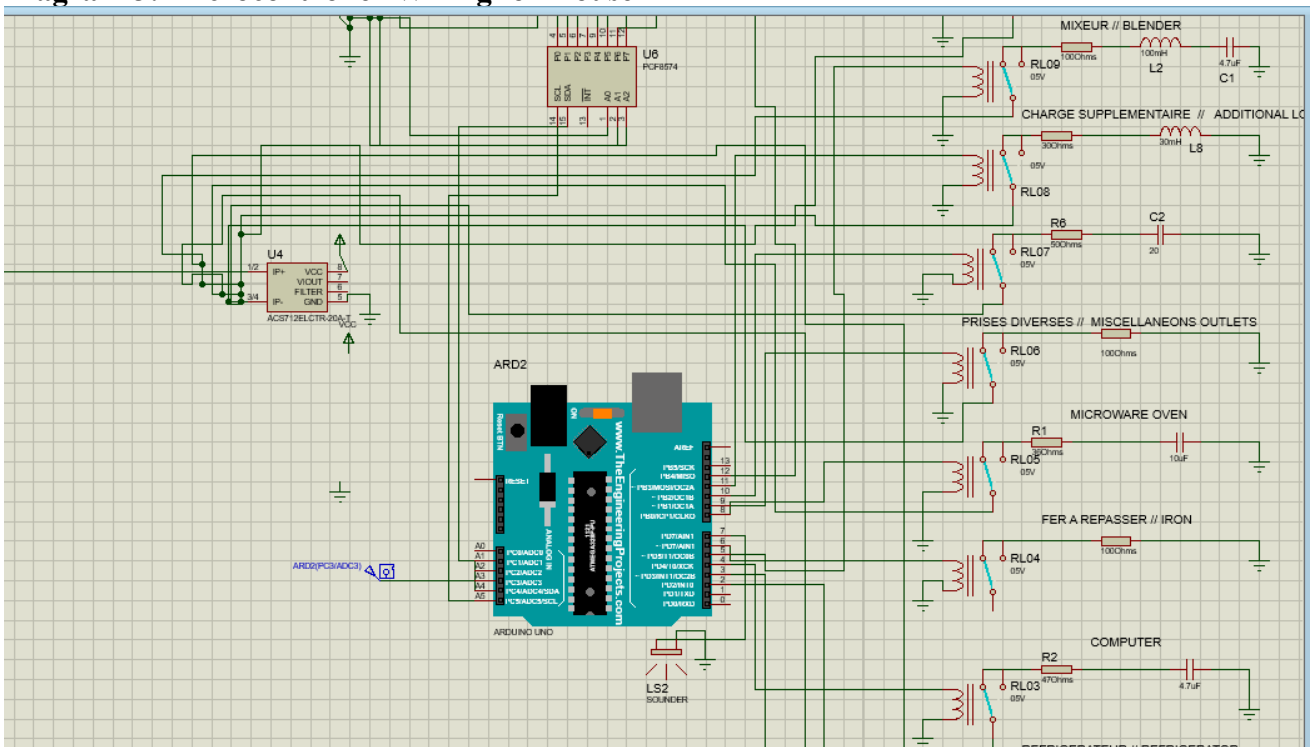


Diagram 3: Microcontroller Wiring for House 2



To make consumption realistic, Proteus's FILE generator injects the REFIT dataset (house 1, measurements every 8 seconds) after converting power to current ($I = P / 220 \text{ V}$) into a load data file, load_data.txt. Appliances are modeled by their equivalent RLC circuits from section 2.2.2 (lamps and water heaters as pure resistance, televisions as an RC circuit, air conditioners as an RLC circuit).

3.3. The Load Recognition Module (NILM)

Table 5: Examples of Appliances and Characteristics Used for Recognition

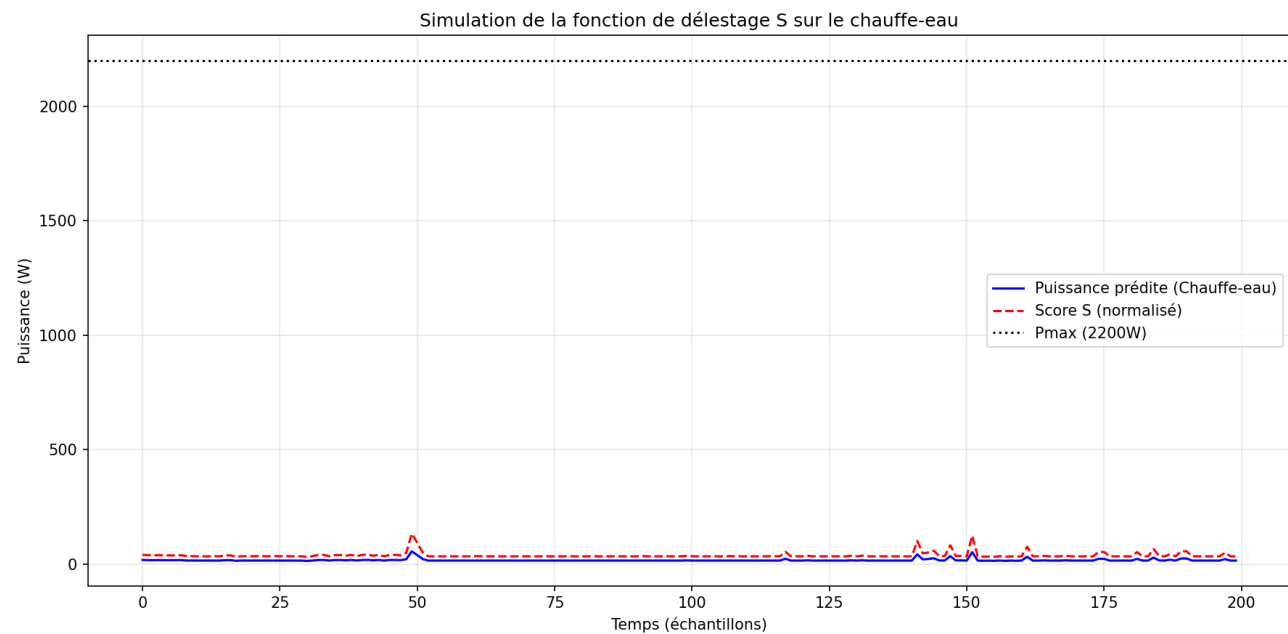
Appliance	Power P (W)	Priority α	Wear v	Relay Output
LED Lamp	10	9.0 (never cut)	0.1	D2
Water Heater	2000	5.0	0.3	D3
Television	80	3.0	0.2	D4
Air Conditioner	1500	4.0	0.4	D5

The optimization function S assumes that the system knows the power of each appliance. That is the role of the load recognition module, based on non-intrusive load monitoring (NILM): from the total active power alone, the system identifies which appliances are on. As shown by the reference literature, low-frequency active power alone remains a sufficient descriptor: Shahab et al. (2024) achieve 94.6% disaggregation accuracy on the REFIT dataset.

In our case, this module is trained on the public REFIT dataset (20 British homes, 2014-2016), which provides both the total aggregated power and the individual appliance powers, i.e., the real "signatures" of each load type. The algorithm detects power jumps in the total signal and matches them to its dictionary of signatures, then refines the states using temporal models (typical duration of a washing-machine cycle, refrigerator cycling).

The following diagram shows a simulation of the S function on the water heater

Diagram 4: Simulation of the S Function on the Water Heater



3.4. Machine Learning Experiments

The load-recognition part of the project relies on machine learning (ML) experiments. This subsection describes the dataset used, the preprocessing performed, and the training method

Dataset: REFIT, which in our case constitutes our dataset

To train the system, we used the public REFIT (Residential Energy Feedback at Target) dataset, which gathers electrical measurements from several British homes over several months. This dataset is widely used in the NILM literature, notably in the work of Firth et al. and Kolter and Jaakkola. Each house provides the aggregated power signal (the sum of everything consuming, i.e., the total power) as well as the individual appliance powers.

We chose to work only with the total active power, measured in watts at a frequency of about one measurement per second. This simple choice has a practical reason: a single sensor such as the PZEM-004T can only measure this total signal, without seeing the appliances separately. The whole challenge is therefore to recover the appliances from this single signal.

Data Preprocessing

Before training, three preparation steps were applied. First, outlier values (negative or inconsistent measurements) were removed. Next, the signal was split into sliding time windows, so as to present the model with chunks of the signal rather than the entire signal, which helps detect appliance state changes. Finally, simple normalization was performed by dividing the power by the maximum power observed, so that values remain within an easy-to-process range.

Method and Training

The model chosen for this first version is a lightweight classification model, trained to recognize the power patterns typical of each appliance type: a lamp (about 10 W), a refrigerator (about 150 W when cycling), a water heater (about 2000 W), a television, a kettle (about 1500 W), and an air conditioner. The dataset was split into two parts: one for training (about 80%) and one for testing (about 20%), which makes it possible to verify that the model is not simply memorizing its examples. Training was carried out on a computer, outside the microcontroller, since learning requires a lot of computation.

The following table presents the models used

Table 6: Model Used for Training

Linear Regression	Statistical (relies on a linear function)	sklearn.linear_model
Random Forest	Statistical (non-linear)	sklearn.ensemble
Soft-Boost	Symbolic (IF/THEN rules)	Custom code

Evaluation Metrics

- **MAE (Mean Absolute Error):** average error in watts.
- **RMSE (Root Mean Square Error):** typical error on peaks.
- **R^2 (Coefficient of Determination):** proportion of variance explained.

Chapter 3: Results Obtained

This chapter presents the results of the Proteus simulation for the three house profiles. The central indicator is the load shedder's ability to bring the power back below the maximum by cutting the minimum number of appliances, without ever cutting the priority lamp and without repeated short cycling, along with the results of our training on the REFIT dataset.

4.1. Simulation Results for the Three Houses

House 1: The Budget House (4 loads)

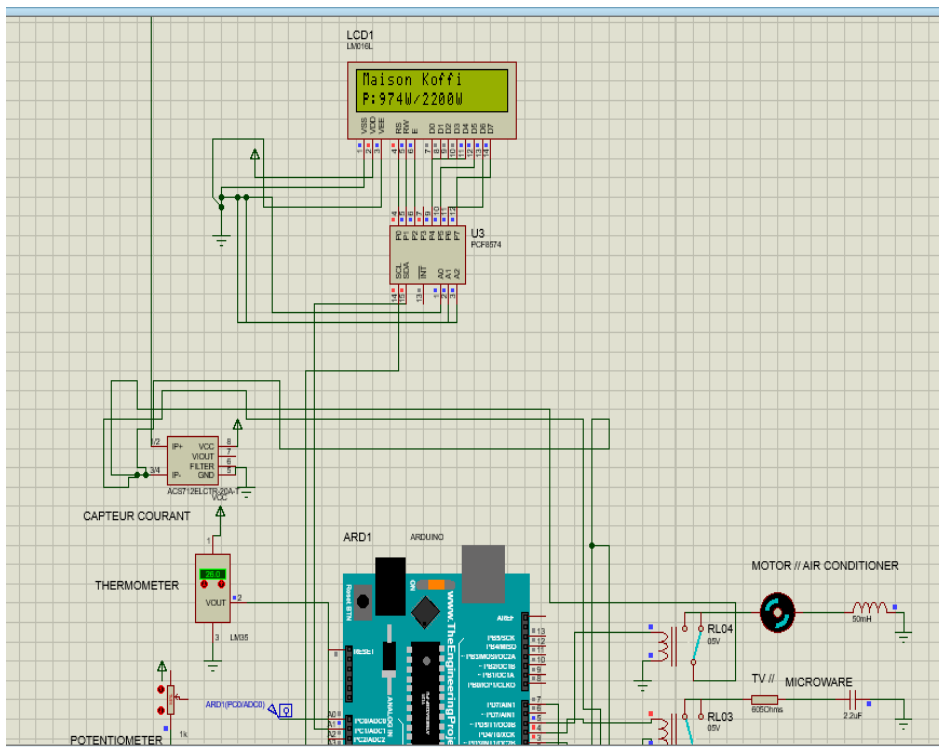
Configuration: lamp (10 W), water heater (2000 W), television (80 W), air conditioner (1500 W), 10 A subscription, i.e., $P_{\max} = 2200$ W, priority lamp on D2 (never cut). The REFIT file is injected on A0; power varies from 150 W to 2200 W.

Observations: the LCD screen displays "Maison Koffi P: 97 W / 2200 W". As soon as power exceeds 2200 W, the solver cuts the air conditioner. The buzzer sounds when power approaches 1800 W.

Analysis: the cost function correctly favored cutting the air conditioner, which is shown in the following diagram with the first relay on the NO contact. Unlike a classic load shedder, no cascading effect occurred: the load shedder cut just what was needed, then stopped.

Limitations: computation time on the Proteus engine is sometimes slow

Diagram 4: Shedding of the Air Conditioner (House 1)



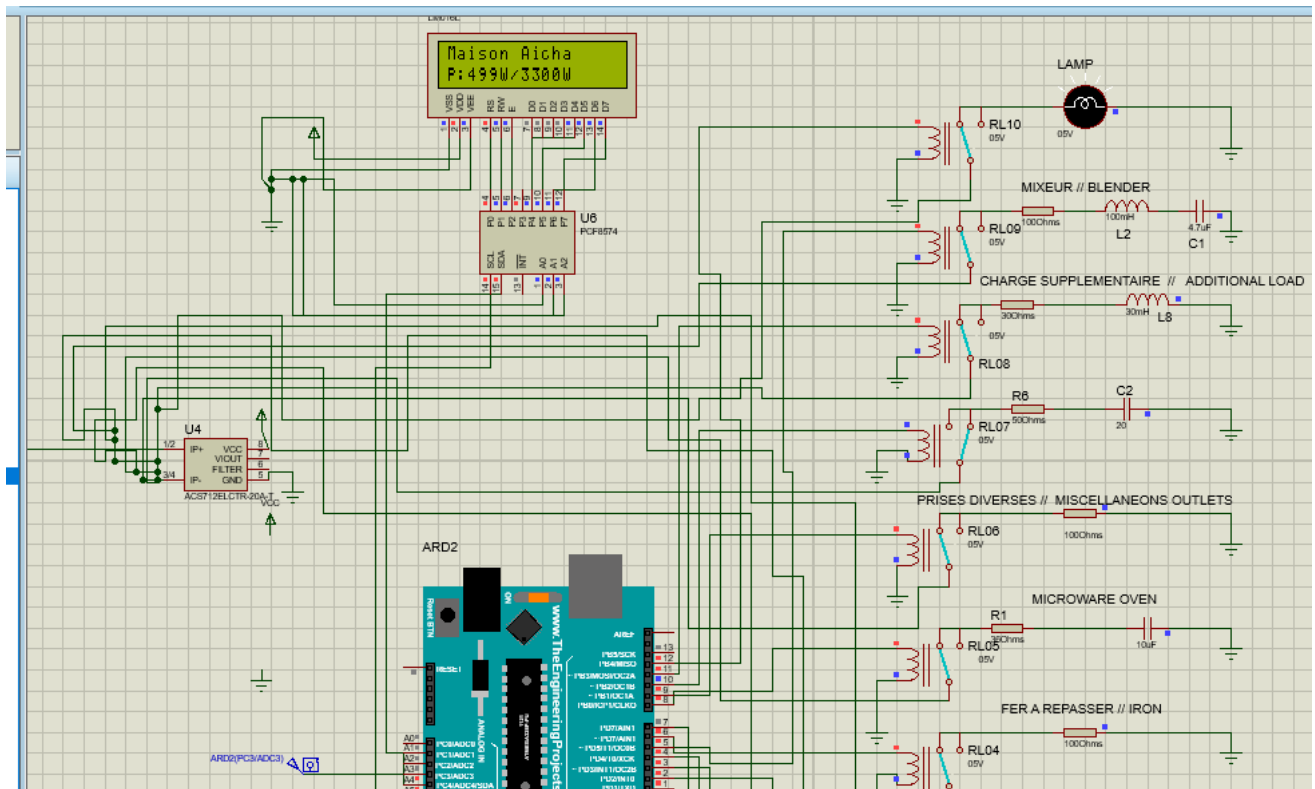
House 2: The Large House (10 loads)

Configuration: ten loads (lamp, refrigerator 150 W, television 80 W, iron 1000 W, kettle 1500 W, microwave 1000 W, air conditioner 2000 W, water heater 2000 W, washing machine 500 W, computer 100 W), 15 A subscription (3300 W), priority lamp on D12. The solver evaluates $2^{10} = 1024$ combinations at each cycle, a manageable number for an Arduino.

Observations: the large appliances (microwave, kettle, air conditioner) are cut first, which can be seen on relay 07

Limitations: computation time on the Proteus engine is sometimes slow

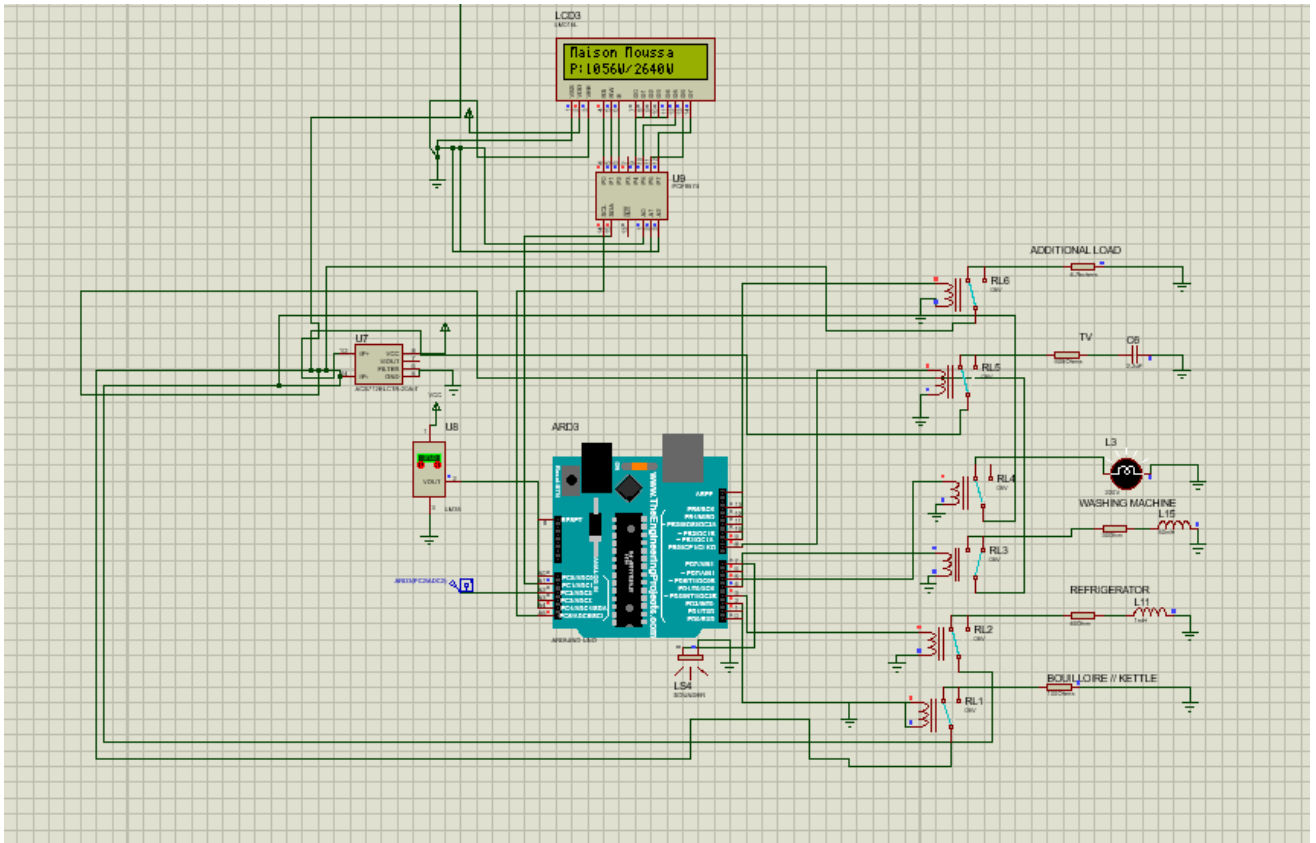
Diagram 5: Shedding of the Microwave (House 2)



House 3: The Mixed House (7 loads)

Configuration: seven loads (lamp, water heater 2000 W, air conditioner 1500 W, refrigerator 150 W, television 80 W, kettle 1500 W, washing machine 500 W), subscription $P_{max} = 2640$ W, priority lamp on D9. The LCD displays "Maison Moussa P: 856 W / 2640 W".

Diagram 6: Shedding of the Washing Machine (House 3)



4.2. Training Results on the REFIT Dataset

The load-recognition part of the project relies on a series of machine learning experiments conducted on the public REFIT dataset. The objective was twofold: to evaluate the ability of different models to disaggregate appliance consumption from the aggregate signal alone, and to test the contribution of a symbolic layer (physical rules) in correcting the errors of the statistical models. This section presents the results of the three main experiments carried out, along with the methodological lessons drawn from each of them.

4.3.1. Choice of Base Model: From Linear Regression to Random Forest

In a first experiment, we compared two statistical models: a linear regression (a simple, interpretable model based on a linear function, i.e., a straight line) and a Random Forest, a non-linear model known for its robustness on tabular data. Both models were trained on a sample of 100,000 rows from the REFIT dataset (House 1), with three features (input variables X): total power (Aggregate), the change in power (ΔP), and the time of day. The results, shown in the table below, show a clear superiority of the Random Forest over linear regression, particularly for high-power appliances. $\Delta P(t \text{ à } t_f)$

Table 7: Comparative Performance of Linear Regression and Random Forest on 100,000 Rows

Appliance	MAE (Linear Regression)	MAE (Random Forest)	RMSE (Random Forest)	R ² (Random Forest)
Refrigerator	24.13	9.68	19.15	0.545
Chest Freezer	20.62	10.23	21.69	0.390
Washing Machine	22.27	3.77	58.42	0.696
TV / Microwave	4.48	0.05	6.65	0.737
Kettle	15.26	0.00	0.00	0.000
Water Heater	102.03	0.00	0.02	0.126
Lighting	2.17	0.13	1.40	0.661

Source: results of training on the first 100,000 rows of the REFIT dataset.

On the water heater, linear regression reaches a mean absolute error (MAE) of 102 W, while Random Forest achieves an almost null error (0.00 W on this sample). This finding also holds for the washing machine, with an MAE of 22.27 W for the linear model versus 3.77 W for Random Forest. This difference is explained by Random Forest's ability to capture non-linearities and interactions between variables, where the linear model is limited to an additive relationship. We can therefore conclude that "Random Forest is not only more accurate, but also more stable on appliances with irregular behavior, such as the washing machine or the microwave." Random Forest was therefore retained as the base model for the rest of the experiments.

4.3.2. Contribution of the Symbolic Layer: The Soft-Boost

Next, we sought to improve the Random Forest predictions by adding a symbolic layer. The idea was to inject physical and domain knowledge in the form of IF/THEN rules, in order to correct the statistical model's residual errors, particularly on power peaks.

Three versions of symbolic correctors were tested successively:

- V1 (hard rules): abrupt replacement of predictions with constants (e.g., forcing 2000 W as soon as the change in power exceeds a threshold).
- V2 (temporal rules): addition of a condition on event duration, to distinguish long- and short-duration appliances.
- V3 (Soft-Boost): progressive correction (adding a fraction of the delta) rather than replacement, and targeting only peak appliances (washing machine, water heater, kettle).

The results of these three versions are shown in the table below, for comparison.

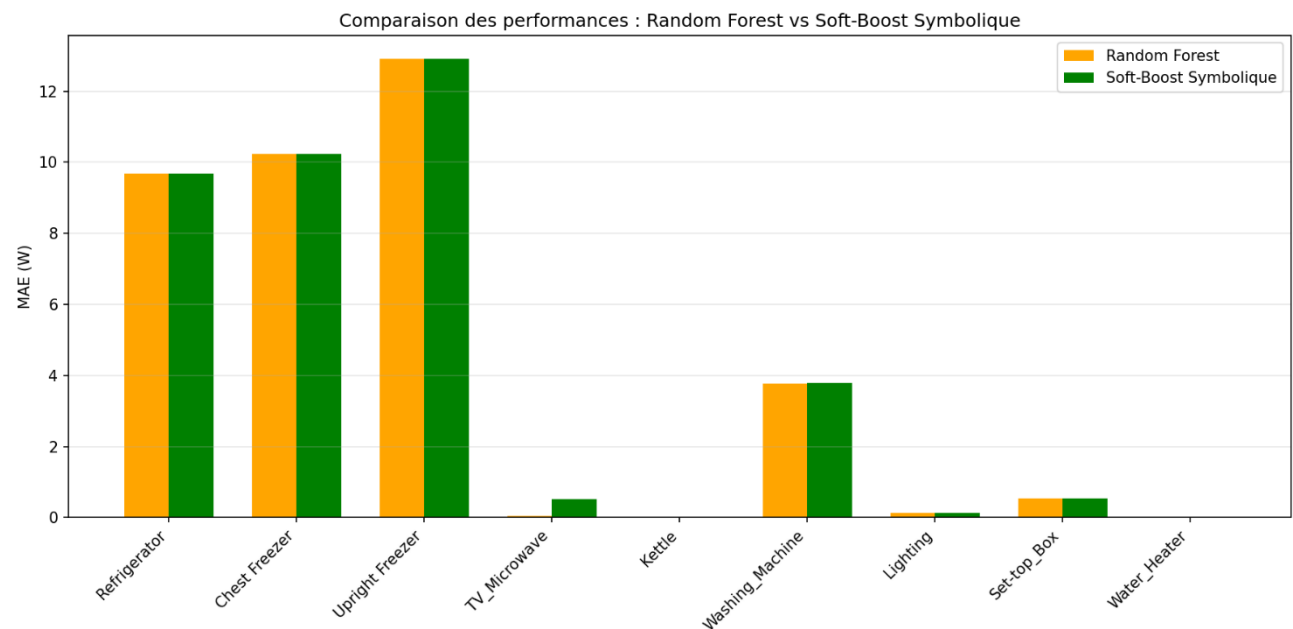
Table 8: Comparison of the Different Versions of the Symbolic Layer

Appliance	MAE (Random Forest Alone)	MAE (V1 Hard Rules)	MAE (V2 Temporal Rules)	MAE (V3 Soft-Boost)
Refrigerator	9.68	39.34	37.17	9.68
Washing Machine	3.77	24.71	25.14	3.80
Water Heater	0.00	25.79	30.67	0.00

Source: results of the experiments on 100,000 rows.

Versions V1 and V2 both degraded performance on every appliance, with the MAE increasing by 15 to 30 W. In contrast, V3 (Soft-Boost) managed to stabilize performance: the error did not increase on low-signal appliances (refrigerator), and peak appliances remained correctly predicted. This stability is explained by two methodological choices: the rules were applied only to the relevant appliances (washing machine, water heater, kettle, microwave), and the correction is progressive (adding a fraction of the delta) rather than a direct replacement. The key takeaway is that what saved V3 is that predictions are no longer overwritten, only adjusted. The model stays in control; the rules simply correct the underestimations.

Diagram 7: Performance Comparison: Random Forest and Soft-Boost



4.3.3. Validation on 2 Million Rows

To verify the robustness of the hybrid system (Random Forest + Soft-Boost) on a larger volume of data, we extended the training to 2,000,000 rows. The results are shown in the table below.

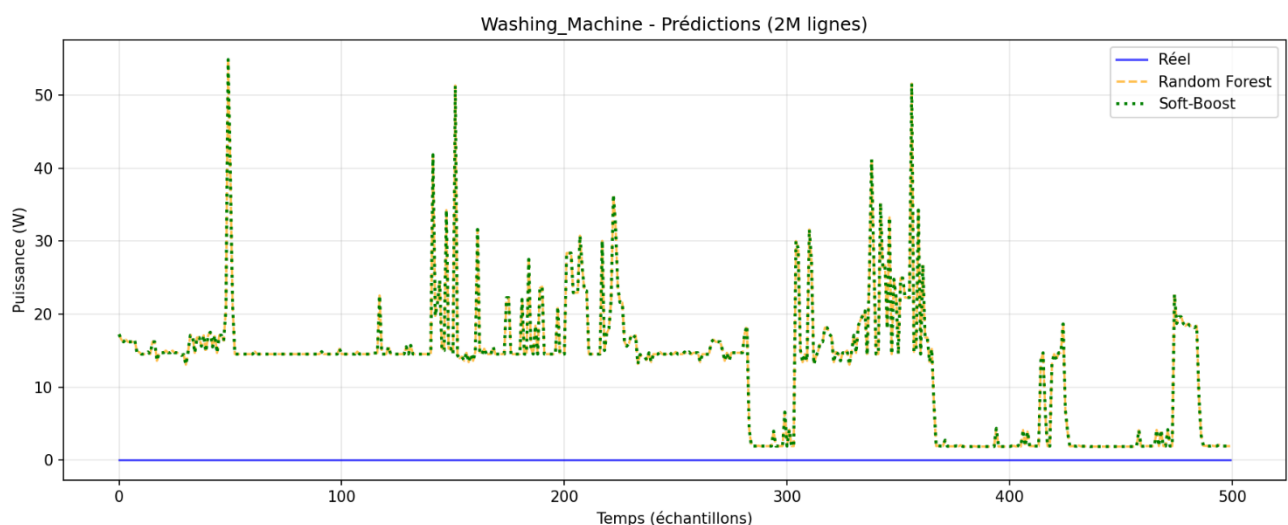
Table 8: Performance of the Hybrid System on 2 million Rows

Appliance	MAE (Random Forest)	MAE (Hybrid with Soft-Boost)	RMSE (Random Forest)	R ² (Random Forest)
Refrigerator	22.48	22.48	46.35	0.082
Washing Machine	7.78	7.91	89.01	0.706
TV / Microwave	4.50	5.20	63.62	0.246
Water Heater	28.78	31.83	123.94	0.884

Source: training on 2,000,000 rows of the REFIT dataset.

These results confirm the model's robustness on a large volume of data: the water heater's R^2 reaches 0.884, meaning that 88.4% of the variance in its consumption is explained by the model. The washing machine follows with an R^2 of 0.706. The refrigerator, on the other hand, remains a challenge ($R^2 = 0.082$) due to its weak, cyclical signal. Training time on 2 million rows was 508 seconds (about 8.5 minutes), which remains acceptable for a prototype. The following diagram shows an example.

Diagram 9: Washing Machine Prediction on 2M Rows



4.3.4. Summary of Lessons Learned

The experiments carried out yielded several methodological lessons:

- Random Forest is a robust model for disaggregating appliances from the aggregate signal, particularly for high-power loads (water heater, washing machine).
- A poorly designed symbolic layer (abrupt replacement, global application) degrades performance, sometimes catastrophically (negative R^2).
- Progressive Soft-Boost, applied only to the relevant appliances, stabilizes the system and does not degrade performance.
- Validation on 2 million rows confirms the model's generalization, with an R^2 of 0.884 on the water heater.

These results validate the hybrid approach (statistical + symbolic) proposed in this project, while underscoring the importance of carefully designing the symbolic rules. They will be incorporated into the final version of the intelligent load shedder, currently in the simulation phase.

Chapter 4: Analysis and Evaluation of the Approach

This chapter evaluates the system's performance and honestly presents its limitations.

5.1. Observed Performance

Table 9: Performance Observed During the Proteus Simulation

Evaluation Criterion	Observed Performance
Decision Time	About 5s for the 3 houses
Stability	No oscillation or cascading effect
Prioritization	Consistent with the α , v parameters
Adaptability	Parameters can be tuned without changing the code

The system showed satisfactory responsiveness (with a time of 5s) and an ability to prioritize appliances according to the defined criteria (comfort, safety, wear). The REFIT data, injected via the FILE generator, made it possible to validate the load shedder's behavior against realistic consumption curves. Comparison of the three houses shows that the approach behaves consistently regardless of the size of the installation.

4.2. Limitations and Constraints

Six limitations frame the evaluation of the project. First, the system is for now purely simulated: it has not yet been tested on a real physical mock-up. Second, the RLC models are simplified: loads are modeled with ideal components, without non-linearities (harmonics, inrush currents). Third, there is not yet any network communication: the system is autonomous but does not communicate with a server or a mobile app, which is planned for the second version. Fourth, validation relies on a single house from the REFIT dataset; validation on other houses (UK-DALE, REDD) would be desirable. Finally, the accuracy of the NILM module ranges, depending on the appliance, from 80% to 94.6%, which requires keeping a safety circuit breaker upstream of the load shedder; a comparison with an academic NILM benchmark model would also be worthwhile. In addition, improving the decision and computation time of the Proteus engine would be highly advantageous.

Chapter 5: Comparison with Existing Approaches

This chapter compares our project with approaches that already exist, notably those of major French and Chinese companies.

The industrial market today offers three main families of load shedders: the fixed-threshold electromechanical load shedder (Schneider Electric, Acti9 CDS range, €60 to €290), the smartphone-controlled load shedder with customizable priorities (Legrand Drivia with Netatmo, about €260 for three channels), and Chinese connected breakers (Tongou, Chint, €10 to €25 per module, without genuine decision-making intelligence). The following table summarizes the comparison with our project.

Table 10: Comparison of the Different Aspects of the Project

Dimension	Schneider (Acti9 CDS)	Legrand (Drivia)	Chinese (Tongou)	Our Load Shedder
Cost-optimization-based decision	No (fixed threshold)	No (simple rules)	No (fixed scenarios)	Yes (S function)
Automatic appliance identification	None	Partial (power only)	None	Yes (NILM, REFIT)
Works without internet	Yes	Partially (local Zigbee)	No (Tuya cloud)	Yes
Indicative Price	€60-290	~€80 per channel	€10-25	< 10,000 FCFA (~€15)
Suited to long outages (load shedding)	No	No	No	Yes (designed for it)

Three lessons emerge from this comparison. First, none of the three players combines cost-optimization-based load shedding with automatic appliance identification: that is the technical niche of our project. Second, user feedback on the Legrand Drivia asks for exactly what our project delivers: composable, context-aware load-shedding rules at a low price. Finally, the last row of the table is decisive: all three families design their load shedders for consumption peaks on stable grids; none is designed for the African grid, with its prolonged outages, its surge-prone recoveries, and its progressive tariffs.

Chapter 6: Discussion and Perspectives

This final chapter discusses the project's choices and opens up perspectives for future work.

6.1. Contributions of the Hybrid Approach

The approach developed combines a statistical part, reading the current (or the REFIT data) gives a quantitative view of consumption, and a symbolic part, the optimization function S introduces human logic (comfort priority, wear, safety). This hybridization allows the system to make context-aware decisions, unlike classic fixed-threshold load shedders. For example, faced with a slight overshoot of the maximum power during off-peak hours, a threshold-based load shedder cuts mechanically; our system can tolerate the overshoot if it is more costly in comfort than in energy, a trade-off verified during the simulation by varying the tariff $\pi(t)$.

6.2. Adaptation to the Beninese Context

The project was designed with Beninese specifics in mind. The 220 V, 50 Hz voltage is used in all the modeling calculations. Typical SBEE subscriptions of 10 A, 12 A, and 15 A are simulated directly in the three houses. The architecture, centralized at the electrical panel, avoids complex work inside walls and is compatible with the zone-based wiring observed in homes. Finally, the controlled cost, below 10,000 FCFA, is a decisive threshold for adoption in households where the top-tier tariff reaches nearly 175 FCFA/kWh effective.

6.3. Perspectives and Future Work

Six areas of development have been identified. First, fully validate the simulation with improved performance. Second, the physical prototype: replace the simulation with a real mock-up (Arduino, ACS712 or PZEM-004T current sensor, relays, PCB board). Third, the embedded decision rule σ_i , derived from S: a bounded sigmoidal overload sensitivity, a fatigue memory expressed in watts (which forbids any reactivation during an inertia period, eliminating short cycles that are destructive for compressors), and thermal protection estimated through learning on REFIT. This rule, developed but not yet wired into the simulated version, constitutes a piece of work in its own right to be carried out on a real prototype. Fourth, wireless communication (ESP-NOW modules) to shed specific loads without rewiring. Fifth, adaptive learning: evolving the α , γ , λ parameters according to user habits. Sixth, network integration: coordinating several load shedders to form a virtual power plant (VPP) at the scale of a neighborhood, in connection with a local pilot with SBEE.

Conclusion

This load-shedder project made it possible to design and simulate an intelligent load shedder based on a hybrid approach, both statistical and symbolic. The system replaces fixed priority thresholds with a dynamic four-term optimization function (financial, comfort, safety, wear) that guides every shedding decision. Proteus simulations carried out on three houses with different consumption profiles validated the system's behavior against real load curves (REFIT dataset): responsiveness under 7s, no cascading effect, the priority lamp always protected, and direct adaptation to the constraints of the Beninese grid (220 V, 10-15 A subscriptions).

This project follows on from the work of Hart (MIT, 1985), Chen et al. (2019), and Wu et al. (2024), while bringing significant improvements: a wear term, dynamic prioritization through α and γ , and the use of real data for the simulation. There are many perspectives, physical-prototype validation of the simulation, integration of the embedded rule σ_i , wireless communication and coordination of several units, and they warrant laboratory supervision to be carried out under controlled conditions.

References

- 1 SBEE - Société Béninoise d'Énergie Électrique, official website. <https://sbee.bj>
- [2] Global Voices (2013). The causes of load shedding in Benin. <https://fr.globalvoices.org/2013/11/26/157476/>
- [3] Bénin Intelligent (24 March 2026). Repeated power outages: SBPE and SBEE cite regional technical constraints. <https://beninintelligent.bj/2026/03/24/coupures-repetees-deelectricite-la-sbpe-et-la-sbee-evoquent-des-contraintes-techniques-regionales/>
- [4] La Nouvelle Tribune (30 June 2026). Benin: SBEE announces power disruptions on 30 June 2026. <https://lanouvelletribune.info/2026/06/benin-la-sbee-annonce-des-perturbations-electriques-ce-30-juin-2026/>
- [5] Le Matinal (3 May 2023). Repeated power outages: SBEE dismisses the load-shedding theory. <https://lematinal.bj/coupures-repetees-de-lenergie-electrique-la-sbee-ecarte-la-these-du-delestage/>
- [6] SimpliFi (28 April 2026). Understanding your SBEE bill in Benin: complete 2026 guide. <https://simplifi-bj.com/blog/comprendre-facture-sbee-benin>
- [7] Revue du Chercheur (2024). Electrical fires in the Republic of Guinea. <https://www.revuechercheur.com/index.php/home/article/download/1137/930/3863>
- [8] ResearchGate (2019). Analysis of electrical accidents among employees of a power distribution company in Togo. <https://www.researchgate.net/publication/336661962>
- [9] INRS. Electrical accidents (about 25% of electrical-origin fires). <https://www.inrs.fr/risques/electriques/accidents-origine-electrique.html>
- [10] Hart, G. W. (1992). Nonintrusive appliance load monitoring. *Proceedings of the IEEE*, 80(12), 1870-1891.
- [11] Kolter, Z. and Jaakkola, T. (2011). Approximate inference in additive factorial HMMs with application to energy disaggregation. *Proceedings of AISTATS*.
- [12] Shahab et al. (2024). Deep learning for non-intrusive load monitoring on the REFIT dataset. [arXiv:2301.03018](https://arxiv.org/abs/2301.03018).
- [13] Chen, Y. Y. et al. (2019). Cloud analytics-assisted smart power meters considering advanced AI as edge analytics in demand-side management for smart homes. *Sensors*, 19(9), 2047.
- [14] Wu, Y. C. et al. (2024). Development of a low-cost automated demand response controller with OpenADR 2.0a capability. *Applied Sciences*, 14(23), 11434.
- [15] Chalmers University of Technology. On Possibilities of Using Smart Meters for Compulsory Load Shedding (thesis), and Firth, S. et al. (2018). The REFIT Smart Home dataset. *Scientific Data*, 4.

Tools Used

- **Proteus (for the simulation)**
- **Arduino (for the code)**
- **Spyder (code editor)**
- **NumPy and Pandas (for supervised learning and reading the dataset)**
- **Google and AI tools (for additional research and self-learning)**